

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

5x4

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

5x1

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}$$

4x5

$$X^T X \text{ Order} = 4 \times 5 \times 5 \times 4 \\ = 4 \times 4$$

$$X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

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$$(X^T X)^{-1} = \begin{bmatrix} 76.9017 & 18.2148 & -3.0043 & -1.1043 \\ 18.2148 & 4.5863 & -0.07241 & -0.02744 \\ -3.0043 & -0.07241 & 0.1602 & 0.0407 \\ -1.1043 & -0.02744 & 0.0407 & 0.0167 \end{bmatrix}$$

(4x5x5x1)

$$X^T Y (4 \times 1) = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T Y = \begin{bmatrix} -99.9112 \\ 1.2174 \\ 2.4166 \\ 0.9058 \end{bmatrix} \quad \theta = \begin{bmatrix} c \\ m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

$$c = -99.9112 \quad m_1 = 1.2174 \quad m_2 = 2.4166 \quad m_3 = 0.9058$$

$$y = 1.2174x_1 + 2.4166x_2 + 0.9058x_3 - 99.9112$$

$$\text{Substitute } x_1 = 9 \quad x_2 = 11 \quad x_3 = 100$$

$$y = 28.208 \text{ kW//}$$